이론시험 문제 해설

문제 1 (10점)

문제 1.A

l_1 l_2 .

(l_1<l_2). $l_2 - l_1 = \frac{L}{2}$ $l_1 + l_2 + \pi R = L,$ $l_2 = \frac{3}{4}L - \frac{\pi}{2}R.$

1. Δt .

$h = l_2 - l_1 = \frac{L}{2}$,

Δx .

.

.

Δx h :

$$\Delta\frac{mv^2}{2} = \frac{m}{L}\Delta x \cdot gh,$$
$$\frac{m2v\Delta v}{2} = \frac{m}{L}v\Delta t \cdot gh.$$

$$a = \frac{\Delta v}{\Delta t} = \frac{h}{L}g = \frac{g}{2}.$$

2. 1 ,

. $\Delta t.$

$l = l_2 + \frac{\pi}{2}R = \frac{3}{4}L$, $M = m\frac{l}{L} = \frac{3}{4}m$, T .

$H = l_2 + R = \frac{3}{4}L - R\left(\frac{\pi}{2} - 1\right)$

. :

$$\Delta\frac{Mv^2}{2} = \frac{m}{L}\Delta x \cdot gH - T\Delta x,$$
$$Mva = \frac{m}{L}v \cdot gH - Tv,$$
$$T = \frac{mgH}{L} - Ma = mg\left[\frac{3}{8} + \frac{R}{L}\left(1 - \frac{\pi}{2}\right)\right].$$

3.

$$\Delta m \cdot a = \Delta m \cdot g\sin\alpha + T_1 - T_2.$$
$T_1 = T_2$,

:

$$\sin\alpha = \frac{a}{g} = \frac{1}{2}.$$
$$\alpha = 30°$$

채점 기준: 문제 1.A

№	평가 항목	
1.		1
2.		1
3.	T_{top}	1
4.	T_{top}	0,5
5.	α	1
6.	α	0,5
	합계	**5,0**

문제 1.B 도우미 증기